GG-S-1616A
17 August 1987
(SUPERSEDING)
GG-S-001616(DSA-DM)
SEPTEMBER 1, 1970

FEDERAL SPECIFICATION

SCISSORS, COLLAR AND CROWN

This specification is approved by the Commissioner Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers scissors suitable for use in cutting bands, preformed crowns, and matrices in dental operative procedures.

1.2 Classification. Scissors shall be of the following styles as specified (see 6.1):

Style I. Curved, Beebee

Style II. Straight, Universal

Style III. Curved, Quimby

2. APPLICABLE DOCUMENTS

2.1. Specifications and standards. The following specifications, standards and drawings, of the issue in effect on date of invitations for bid or requests for proposal, form a part of this specification to the extent specified herein.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RSTS, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Federal Specifications

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

Federal Standards

FED-STD-123	- Marking for Domestic Shipment (Civilian Agencies).
FED-STD-151	- Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402).

(Copies of listed federal and military standards, specifications and Commercial Item Descriptions (CIDs), listed in the Department of Defense Index of Specifications and Standards (DoDISS), should be obtained from the DoD Single Stock Point, Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. Copies of industry association documents should be obtained from the sponsor. Copies of all other listed documents should be obtained from the contracting activity or as directed by the contracting activity).

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies).

Military Specifications

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.
MIL-P-116	Methods of Preservation.

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Military Standards

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packing of.

DRAWINGS

Defense Personnel Support Center (DPSC)

20178	Scissors, Collar and Crown.
23201	Scissors, Collar and Crown, Curved, Quimby.

(Copies of Military specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bid or requests for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E18	Tests for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
E92	Tests for Vickers Hardness of Metallic Materials.
F899	Stainless Steel Billet, Bar and Wire for Surgical Instruments.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification however supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General requirements. The scissors shall be in accordance with the applicable requirements for class 4, martensitic stainless steel of GG-I-526, except for material (see 3.2), and with the additional options and requirements specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are specification control drawings. The supplier shall be responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawing to insure proper design and function of the item. No deviation from the prescribed dimensions and tolerances is permissible without prior approval of the contracting officer.

3.2 Materials. The scissors shall be fabricated of class 4, type 420B, 420X, 420C or 420F stainless steel, as specified in ASTM F899. Except, the screw shall be fabricated of class 4, type 416 stainless steel.

3.3 Hardness. After proper stress relieving, styles I, II, and III scissors, shall have a Rockwell hardness of C50 to C55 (with the exception of the screw). Hardness shall be determined as specified in 4.4.2.

3.4 Style, design, and dimensions. Scissors shall be suitable to contour preformed crowns, and to cut dental wires and matrices.

3.4.1 Style I scissors. Style I shall be a curved blade Beebee design in accordance with DPSC Drawing 20178.

3.4.2 Style II scissors. Style II shall be a sturdy, straight blade, Universal design in accordance with DPSC Drawing 20178.

3.4.3 Style III scissors. Style III shall be a curved Quimby design, with double pointed ends, in accordance with DPSC Drawing 23201.

3.5 Construction. The scissors shall be of three piece construction consisting of two scissor sections locked together by a screw, which is permanently set in a fixed position. The screw lock and scissor sections shall be properly assembled to provide a cutting portion which shall be free of both excessive stiffness and looseness.

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The inside of the cutting portion of each scissor shall be ground to properly mate with the opposing scissor during cutting procedures, and the cutting edges of the blades shall be properly sharpened.

3.6 Finish. The scissors shall have a mirror finish, except for the ground inside portion of the blades which shall have a crocus finish as specified in GG-I-526.

3.7 Corrosion resistance. Styles I, II, and III scissors, shall meet the boil and copper sulfate test requirements of GG-I-526.

3.8 Identification markings. Instruments of foreign manufacture shall also be marked in addition to the requirements of GG-I-526, with the word "stainless."

3.9 Cutting performance. The scissors, style I and style III, shall cut a continuous length, along the blade to the tip; style II shall cut stainless steel wire, when tested as specified in 4.4.3. In addition, style III scissors shall also properly cut a copper band when tested as specified in 4.4.3.

3.10 Workmanship. Each pair of scissors shall be free from defects which detract from its appearance or impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the scissors, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in tables II and IV (see 4.2.1 and 4.3) and table III (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table II. Unit of product for examination purposes shall be one pair of scissors.

TABLE II. Examination criteria

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table III. The unit of product for tests purposes shall be one instrument.

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TABLE III. Sampling criteria for tests

Characteristic	Requirement	Test Procedure	Inspection level	AQL (Percent defective)
Material	3.2	4.4.1	See 4.2.3	1/
Hardness	3.3	4.4.2	S-2 2/	1.5
Cutting performance	3.9	4.4.3	II	1/
Corrosion resistance	3.7	GG-I-526	S-2	1.0

1/ Acceptance number shall be zero for all sample sizes.

2/ In lieu of sampling plans for normal inspection of MIL-STD-105, sampling plans for reduced inspection may be used when the inspection lot or batch complies with both of the below listed conditions:

a. Consists of units manufactured from the same melt of metal, specified chemical composition, produced by the same manufacturer at the same plant, and using the same facilities at essentially the same time.

b. Consists of units subjected to the same heat treatment processes simultaneously in the same equipment; except that when continuous belt automatic heat treatment equipment is used, a heat treatment lot shall be considered no more than one day's production under constant conditions.

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the construction of the scissors shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same unit of product shall be used for the determination of two or more test characteristics.

4.3 Examination. The scissors shall be examined for defects including, but not limited to, those listed in table IV.

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TABLE IV. Classification of defects

Category	Defect
Major A	
101	Style and design not as specified for style of instrument.
102	Cutting portion of scissors not free from excessive stiffness or looseness.
103	Screw in lock not permanently set in a fixed position.
104	Cutting edges not free from defects such as nicks, feathers, or burrs.
105	Scissors not free from defects such as cracks, pits, or corroded areas.
Major B:	
151	Operative end dimensions not as specified.
Minor A:	
201	Finish not highly polished, or ground portion of blades not crocus finished.
202	Finish not free from coarse grind marks or
203	Scissors not free from discoloration. scratched surface areas.
204	Identification marking not permanent, legible, or complete.
205	Packaging or packing not as specified.
206	Marking on packaging or pack not as specified.
Minor B:	
251	Dimensions (other than 151) not as specified.

4.4 Tests procedures.

4.4.1 Chemical analysis. The chemical analysis shall be conducted in accordance with methods 111 or 112 of FED-STD-151. In the event of conflict, method 111 shall govern.

4.4.2 Hardness. Hardness shall be determined in accordance with ASTM Designation E18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

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4.4.3 Cutting performance. Test materials shall be a strip of austenitic stainless steel 0.002 ± .0005 inch thick, with a Rockwell hardness of C35 to C40, and a surgical stainless steel wire, 0.020 inch nominal. Scissors to be tested shall be thoroughly cleaned prior to testing. Three separate, consecutive tests shall be performed on each material for style I, style II, and style III. Each test shall consist of making a 1/2 inch (minimum length) cut for style I and style III scissors, and a cut through the stainless steel wire for style II scissors. The cuts shall be made using the scissors in the normal manner, with the fingers in the finger rings of the scissors, except that no lateral pressure shall be used. In addition, style III scissors shall be provided with blades of a width, thickness, and curvature that permit precise gingival festooning on an annealed No. 1 copper band, 0.189 ± 0.005 inch inside diameter and 0.005 ± 0.0015 inch wall thickness.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Scissors shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each scissors shall be wrapped in neutral tissue or other suitable anticorrosive material. Each scissors shall be packaged in accordance with MIL-P-116, submethod IC-2. As an alternate, each scissors shall be packaged in a rigid plastic container having a suitable closure. Closure shall be as specified in the applicable box specification.

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5.1.1.4 Intermediate package. Twelve unit packages containing scissors of one style only shall be packaged in a box constructed in accordance with PPP-B-566, PPP-B-636, class domestic, or PPP-B-676. Closure shall be as specified in the applicable box specification. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.2 Level C. The level C preservation for the scissors shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Twenty-four intermediate packages (288 units) containing scissors of one style only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Twenty-four intermediate packages (288 units) containing scissors of one style only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for the scissors shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container; such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

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5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceed 250 sounds or 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 54 inches in height, 48 inches in width, and 40 inches in length. Pallet shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. The scissors packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. The scissors packed as specified in 5.2.3 shall be unitized as specified in 5.2.5.

5.3 Marking.

5.3.1 Levels A, B, and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Style required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-00-118-6250	SCISSORS, COLLAR AND CROWN, Curved, Quimby, 4-3/4 inch (12.06 cm).
6520-00-554-8140	SCISSORS, COLLAR AND CROWN, Curved, Beebee, 4-1/4 inch (10.79 cm).
6520-00-554-8150	SCISSORS, COLLAR AND CROWN, Straight, Universal, 4-1/2 inch (10-16 cm).

6.3 Subject term (key word) listing:
Beebee
Collar and Crown
Curved Scissors
Quimby
Straight, Universal Scissors

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review Activities:

Army - MD
Navy - MS
Air Force - 03

Civil Agency Coordinating Activities:
VA-OSS
PHS
FDA-MPQAS

49106
49107
49108

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6520-1903

Location: NCR Disk 6

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GG-S-1616A	SCISSORS, COLLAR AND CROWN

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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